

Supplementary Material for Physics-Informed State Reconstruction and Event-Source Inference From Sparse PMUs Under Limited Observability

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ABSTRACT This supplement records the detailed literature matrix, electrical conventions, mathematical proofs, Fisher-information analysis, Bayesian estimator construction, frozen experimental contracts, and secondary results supporting the main article. It distinguishes the evaluated state and 137-hypothesis load-support estimators from the still-unexecuted common-contract joint formulation. It does not add empirical claims or change any test denominator. Every numerical table and empirical figure is generated from the same hashed evidence snapshot used by the main manuscript.

INDEX TERMS Bayesian inference, Fisher information, functional observability, phasor measurement units, reproducibility, state estimation, event-source inference.

APPENDIX A DETAILED LITERATURE CAPABILITY MATRIX

Table 1 expands the family-level synthesis in the main paper. “Held source” means that the physical asset is absent from source-labeled training and tuning. The entries describe published tasks and assumptions; they are not a cross-paper performance ranking.

The direct competitors solve different contracts. State and unknown-input estimators define the latent reconstruction problem but not the event taxonomy. Learned localizers rank represented sources but rarely remove the physical asset from every source-labeled fitting operation. Model-based outage methods provide the closest likelihood precedent, while functional observability and placement studies provide the information tests. These distinctions determine the baselines and prevent cross-paper accuracy comparisons with incompatible targets.

APPENDIX B ELECTRICAL MODEL AND EVENT OPERATORS

Within a smooth network mode, the DAE is locally index one after fixing the reference angle. Differentiating the algebraic constraint gives

$$\dot{z} = -g_z^{-1}(g_x f + g_u \dot{u} + g_\theta \dot{\theta}). \quad (1)$$

This reduction is not applied across a topology switch or fault insertion, where algebraic variables may jump. Differential states remain continuous unless a simulator-specific reset is declared.

With $Y_{\text{bus}} = G + jB$ and $V_i = |V_i|e^{j\vartheta_i}$, the network injections are

$$P_i = |V_i| \sum_j |V_j| (G_{ij} \cos \vartheta_{ij} + B_{ij} \sin \vartheta_{ij}), \quad (2)$$

$$Q_i = |V_i| \sum_j |V_j| (G_{ij} \sin \vartheta_{ij} - B_{ij} \cos \vartheta_{ij}). \quad (3)$$

For branch $e = (i, j)$ with complex tap t_e , series admittance y_e , and terminal shunts $y_e^{f,sh}$ and $y_e^{t,sh}$,

$$\begin{bmatrix} I_{e,f} \\ I_{e,t} \end{bmatrix} = \begin{bmatrix} (y_e + y_e^{f,sh})/|t_e|^2 & -y_e/t_e^* \\ -y_e/t_e & y_e + y_e^{t,sh} \end{bmatrix} \begin{bmatrix} V_i \\ V_j \end{bmatrix}. \quad (4)$$

The implemented PMU operator selects the actual terminal and orientation. Replacing this current with complete bus-injection current changes the likelihood.

Positive-sequence phasors define the balanced model. Negative- and zero-sequence components remain integrity or unbalance evidence. Ideal continuous-frequency channels satisfy

$$f_i = f_0 + \frac{1}{2\pi} \frac{d\angle V_i}{dt}, \quad \text{ROCOF}_i = \frac{1}{2\pi} \frac{d^2\angle V_i}{dt^2}, \quad (5)$$

TABLE 1. Detailed capability comparison for the closest methodological works.

Work	State target	Event/source target	Physics mechanism	Sparse sensing	Held source	Integrity treatment
Katanic <i>et al.</i> [1]	Differential and algebraic DAE states	Unknown injectors	Incomplete nonlinear DAE and least squares	Yes	N/A	No
Nugroho <i>et al.</i> [2]	Differential and algebraic DAE states	Unknown input and input-sensor failure	Outlier-resistant DAE observer	PMU voltage/current	N/A	Failure tolerance
Qu <i>et al.</i> [3]	Machine state	Unknown field voltage and torque	Event-triggered nonlinear state/input estimator	Event-triggered PMUs	N/A	Covariance bounds
Ngo <i>et al.</i> [4]	Power-system states	None	Branch-current physics plus graph learning	Yes	N/A	No
Yue <i>et al.</i> [5]	Distribution-system state	Line outage	Grid-model-informed graph learning	Yes	Not stated	Outlier-resistant SE
Moshagh <i>et al.</i> [6]	Transmission-system state	Topology variation	Topology-aware graph learning	PMU-unobservable systems	Not a source task	PMU failure and bad data
Pandey <i>et al.</i> [7]	None	Detection, class, and source	Topology/equipment information	PMU network	Not stated	No
Yildiz and Abur [8]	None	Fault detection and location	OLS/CNN and PMU placement	Yes	Not stated	No
Li <i>et al.</i> [9]	None	Multi-family disturbance source	Topology/type-constrained STGCN loss	Yes	Not stated	No
Kummerow <i>et al.</i> [10]	None	Known/unknown event type and location	Siamese distance and learned rejection	Limited/full sets	Open-set rejection	Rejection boundary
Hu and Cheng [11]	None	Open-set event detection/classification	Generative window evidence and fusion	Streaming PMUs	Unknown-event category	No source posterior
Rimkus <i>et al.</i> [12]	Missing PMU trajectories	Downstream fault localization	Graph reconstruction before diagnosis	Yes	Imputer only	Corruption via imputation
Zhai <i>et al.</i> [13]	Full dynamical state	Line-attack location	Learned temporal dynamics	Yes	Across benchmarks	Attack target only
Main article	Hidden voltage field	All load supports with $K \leq 2$; historical eight-label ontology	DAE hypotheses and quadratic physical likelihood	Eight PMU buses	Physical bank has no source-label head	Physical/integrity variables separate

but an operational likelihood must reproduce the causal filtering of the PMU backend.

APPENDIX C METRICS, CAUSALITY, AND EVALUATION UNITS

For trajectory r , hidden-voltage relative error is

$$\text{HVRE}^{(r)} = \frac{1}{T_r |\mathcal{U}|} \sum_{k=1}^{T_r} \sum_{i \in \mathcal{U}} \frac{|\hat{V}_{i,k} - V_{i,k}|}{\max(|V_{i,k}|, \epsilon)}. \quad (6)$$

It is a trajectory-and-bus mean reported as a percentage; it is not the standard per-frame synchrophasor TVE conformity statistic. Detection, event-family macro-F1, exact physical Top-1/Top-3, cardinality, candidate-set coverage, and integrity-source Top-1 retain their own populations. A joint outcome is correct only when every applicable component is correct.

A causal estimate at k uses $y_{0:k}$. A fixed-lag smoother, complete-window score, or centered temporal feature is retrospective. For source generalization, the physical asset is absent from source-labeled training, feature fitting, threshold selection, and calibration. Intervals and significance tests use the complete trajectory, event parent, or rebuilt plant as the independent unit; samples within a 30-Hz stream are correlated observations, not replicates.

APPENDIX D PROOFS AND ADDITIONAL DIAGNOSABILITY RESULTS

This appendix develops the mathematical chain used by the paper. It begins with the sampled hybrid event map, separates state-functional recovery from event-source separation, derives the relevant Fisher information after physical nuisance is removed, and then states the finite-horizon and temporal limits. The results are local unless a statement explicitly says otherwise.

A. REGULARITY AND STATISTICAL CONTRACT

Let $u = \text{col}(x, z)$ collect differential and algebraic coordinates and consider the descriptor model

$$E\dot{u} = F(u, p), \quad y = h(u, p). \quad (7)$$

The analysis uses four assumptions.

- 1) The reference angle has been fixed and the algebraic Jacobian is nonsingular on the local mode under study, so a regular index-one chart exists between discrete events.
- 2) The scheduled event time is fixed. In a neighborhood of zero amplitude, the complete callback–solver–sampling map is twice continuously differentiable.
- 3) Every covariance used as a metric is positive definite on the retained measurement subspace. Candidate-dependent covariances retain their log determinants in likelihood comparisons.
- 4) The support set is finite and all Bayesian priors are proper. Exact signed-amplitude equivalences use a

TABLE 2. Physical intervention operators represented by the common DAE formulation.

Family	DAE action at source ℓ	Candidate support	Severity coordinates	State continuity
Bus fault	Add a shunt primitive for the clearing interval	admissible buses	complex admittance or resistance/reactance	x continuous; z may jump
Line outage	Remove branch tap, series, charging, and shunt primitives	in-service branches	status and switching time	x continuous; z may jump
Generation change	Modify scheduled injection or a declared governor reference	generator buses	$\Delta P_G, \Delta Q_G$, duration	continuous input-driven response
Load change	Modify the active/reactive withdrawal model	load buses	$\Delta P_L, \Delta Q_L$, ZIP convention	continuous input-driven response

neighborhood of zero within each support subspace; constrained physical amplitudes instead use their actual difference set.

These assumptions distinguish the local theory from general descriptor left invertibility, state-triggered saltation, and global nonlinear identifiability.

B. SIMULATOR-CONSISTENT EVENT SENSITIVITIES

For a fixed-time, impulse-free event, the differential state is continuous, but the algebraic state must satisfy the post-event constraint. This physical right limit must be distinguished from the values stored inside a simulator callback and from the first sampled frame. The following result makes that distinction explicit.

Theorem 1 (Local consistency-jump expansion): Let $x_0 = x(\tau^-)$ and suppose $g_+(x_0, z_0, 0) = \mathbf{0}$. If g_+ is three times continuously differentiable and g_z is nonsingular at $(x_0, z_0, 0)$, there is a unique local consistent branch

$$z_+(a) = z_0 + az_1 + \frac{1}{2}a^2z_2 + O(a^3), \quad (8)$$

where all derivatives below are evaluated at the nominal right-limit mode and

$$z_1 = -g_z^{-1}g_a, \quad (9)$$

$$z_2 = -g_z^{-1}(g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa}). \quad (10)$$

The corresponding differential state satisfies $x(\tau^+) = x(\tau^-)$, and its first event-amplitude sensitivity starts with

$$\dot{x}_a(\tau^+) = f_a - f_z g_z^{-1}g_a \equiv B_{r,a}. \quad (11)$$

Its second-order initial forcing is

$$\begin{aligned} \dot{x}_{aa}(\tau^+) \equiv B_{r,aa} = & f_{zz}[z_1, z_1] + 2f_{za}z_1 + f_{aa} \\ & - f_z g_z^{-1}(g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa}). \end{aligned} \quad (12)$$

a: Proof.

The implicit-function theorem applied to $g_+(x_0, z_+(a), a) = \mathbf{0}$ gives a unique smooth local branch. Differentiating once gives $g_z z_1 + g_a = \mathbf{0}$, which yields (9). Differentiating a second time gives $g_z z_2 + g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa} = \mathbf{0}$, which yields (10). An impulse-free fixed-time switch does not reset the energy-storage coordinates, so the differential state is continuous. Differentiating $\dot{x} = f_+(x, z_+(a), a)$ at the right limit, with $x_a(\tau^+) = \mathbf{0}$, gives (11). A second differentiation, using $x_{aa}(\tau^+) = \mathbf{0}$ and (10), gives (12). $\square$

The theorem also defines an implementation-independent consistency diagnostic. If a callback changes the parameter but temporarily retains z_0 , its algebraic residual is

$$r_{cb}(a) = g_+(x_0, z_0, a) = ag_a + \frac{1}{2}a^2g_{aa} + O(a^3). \quad (13)$$

Substituting the zeroth-, first-, and second-order approximations to (8) produces constraint residuals of orders $O(a)$, $O(a^2)$, and $O(a^3)$, respectively. Central finite differences of an independently solved algebraic branch can therefore test both derivatives and the expected convergence orders without relying on undocumented callback storage.

For a differentiable PMU map $y = h(x, z, a)$, the direct right-limit signature is

$$y_a(\tau^+) = h_a - h_z g_z^{-1}g_a, \quad \Delta y_P = aM_P y_a(\tau^+) + O(a^2). \quad (14)$$

This direct algebraic visibility is not the entire event signature. Even when (14) is nearly zero at the retained PMUs, (11) can generate a distinguishable dynamic response over later frames. Instantaneous invisibility therefore does not imply finite-horizon nonidentifiability.

The frozen artifacts begin at the first retained 30-Hz frame and do not record the callback microstate or an independently solved algebraic right limit. Accordingly, the paper treats the right-limit equations above as the physical contract, not as an observed fact about the current ANDES callback implementation. The relevant differentiable object for the existing response dictionary is the sampled map

$$\Psi_{T_s}^{\text{sim}} : (u^-, p(a)) \mapsto u(\tau + T_s; a), \quad T_s = 1/30 \text{ s}. \quad (15)$$

For event coordinates a_i , define

$$L_i = \left. \frac{\partial \Psi_{T_s}^{\text{sim}}}{\partial a_i} \right|_{a=\mathbf{0}}, \quad L_{ij}^{(2)} = \left. \frac{\partial^2 \Psi_{T_s}^{\text{sim}}}{\partial a_i \partial a_j} \right|_{a=\mathbf{0}}. \quad (16)$$

For $t \geq \tau + T_s$, the first and second sensitivities obey

$$E\dot{s}_i = F_u s_i + F_p p_i, \quad (17)$$

$$E\dot{s}_{ij} = F_u s_{ij} + r_{ij}, \quad (18)$$

where

$$\begin{aligned} r_{ij} = & F_{uu}[s_i, s_j] + F_{up}[s_i, p_j] + F_{pu}[p_i, s_j] \\ & + F_{pp}[p_i, p_j] + F_p p_{ij}. \end{aligned} \quad (19)$$

The initial conditions are $s_i(\tau + T_s) = L_i$ and $s_{ij}(\tau + T_s) = L_{ij}^{(2)}$. For the affine event parameterization, $p_{ij} = \mathbf{0}$.

Theorem 2 (Propagation of event-map mismatch): Suppose two second-order models have the same nominal trajectory,

first-order sensitivities, parameter derivatives, and forcing in (18), but different second-order event-map initial conditions. Their difference $e_{ij} = s_{ij}^A - s_{ij}^B$ satisfies

$$E\dot{e}_{ij} = F_u e_{ij}, \quad e_{ij}(t) = \Phi_E(t, t_1)e_{ij}(t_1) \quad (20)$$

on the consistent solution subspace, where $t_1 = \tau + T_s$.

b: Proof.

Subtract (18) for the two models. Every forcing term cancels by assumption, leaving the homogeneous descriptor variational equation. Applying its transition operator gives (20). $\square$

When the retained PMU operator is locally affine, $y = H_y u + d$, the sampled response coefficients are

$$D_i = H_y s_i, \quad Q_i = \frac{1}{2} H_y s_{ii}, \quad Q_{ij} = H_y s_{ij}. \quad (21)$$

Thus all quadratic curvature comes from the physical solution and event map, not from PMU post-processing. This is why the dictionary must differentiate the same event semantics used to generate the trajectories.

C. EXACT FUNCTIONAL RECONSTRUCTION

Consider the finite-horizon linear experiment

$$Y = \mathcal{O}x + \varepsilon, \quad \psi = Lx, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, R), \quad R \succ \mathbf{0}. \quad (22)$$

Theorem 3 (Functional reconstruction equivalence): The following statements are equivalent: (i) $\mathcal{O}x_1 = \mathcal{O}x_2$ implies $Lx_1 = Lx_2$; (ii) $\ker \mathcal{O} \subseteq \ker L$; and (iii) there is a linear map K such that $L = K\mathcal{O}$.

a: Proof.

If (i) holds and $v \in \ker \mathcal{O}$, the observations of v and $\mathbf{0}$ coincide, hence $Lv = \mathbf{0}$ and (ii) follows. If (ii) holds, define $K(\mathcal{O}x) = Lx$ on $\text{range } \mathcal{O}$. The definition is independent of the representative because two representatives differ by a vector in $\ker \mathcal{O} \subseteq \ker L$. Extend K linearly to the observation space to obtain (iii). Finally, (iii) immediately implies (i). $\square$

Let

$$G = \mathcal{O}^\top R^{-1} \mathcal{O}. \quad (23)$$

Theorem 4 (Best linear unbiased estimate of an estimable function): If the equivalent conditions of Theorem 3 hold, then

$$\hat{\psi}_* = LG^\dagger \mathcal{O}^\top R^{-1} Y, \quad (24)$$

$$\text{Cov}(\hat{\psi}_*) = LG^\dagger L^\top. \quad (25)$$

It has minimum covariance, in the positive-semidefinite order, among all linear unbiased estimators of Lx [14].

b: Proof.

Because $v^\top G v = \|\mathcal{O}v\|_{R^{-1}}^2$, $\ker G = \ker \mathcal{O}$. Estimability gives $L = LG^\dagger G$, so taking expectation in (24) yields Lx . Direct covariance propagation gives (25). Let AY be any other linear unbiased estimator and set $T = A - LG^\dagger \mathcal{O}^\top R^{-1}$.

Then $T\mathcal{O} = \mathbf{0}$ and the cross covariance vanishes because $LG^\dagger \mathcal{O}^\top T^\top = \mathbf{0}$. Consequently,

$$\text{Cov}(AY) = \text{Cov}(\hat{\psi}_*) + TRT^\top \succeq \text{Cov}(\hat{\psi}_*). \quad (26)$$

If the kernel inclusion fails, two states induce the same data law but different targets, so no estimator can be unbiased for every state. $\square$

The pseudoinverse has this interpretation only for estimable targets. In the nonlinear DAE, a pointwise kernel test is not sufficient: local factorization requires the condition to hold on a regular constant-rank neighborhood with connected local fibers [15].

D. FISHER INFORMATION WITH PHYSICAL NUISANCE

For a differentiable Gaussian model

$$r \mid \theta \sim \mathcal{N}(\mu(\theta), \Sigma(\theta)), \quad (27)$$

the Fisher information contains both mean and covariance contributions. For scalar a ,

$$I(a) = \mu_a^\top \Sigma^{-1} \mu_a + \frac{1}{2} \text{tr}(\Sigma^{-1} \Sigma_a \Sigma^{-1} \Sigma_a). \quad (28)$$

The second term must be retained when event amplitude changes the covariance. The evaluated load bank freezes Σ , so only the first term is used.

To expose nuisance geometry, consider

$$r = ad + N\nu + \varepsilon, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, \Sigma), \quad (29)$$

where ν may contain initial-state, operating-point, parameter, or integrity directions. Let $W^\top W = \Sigma^{-1}$, $\bar{d} = Wd$, $\bar{N} = WN$, and

$$P_\perp = I - \bar{N}\bar{N}^\dagger. \quad (30)$$

Theorem 5 (Efficient Fisher information after nuisance profiling): For deterministic unknown ν , the efficient information for a is

$$I_{a|\nu}^{\text{det}} = \bar{d}^\top P_\perp \bar{d} \quad (31)$$

$$= d^\top \Sigma^{-1} d - d^\top \Sigma^{-1} N (N^\top \Sigma^{-1} N)^\dagger N^\top \Sigma^{-1} d. \quad (32)$$

It is zero exactly when the event tangent lies in the observable nuisance span.

a: Proof.

After whitening, the score for a is $\bar{d}^\top \bar{\varepsilon}$. Nuisance scores span $\{v^\top \bar{\varepsilon} : v \in \text{range } \bar{N}\}$. Their Fisher inner product is Euclidean because $\mathbb{E}[\bar{\varepsilon}\bar{\varepsilon}^\top] = I$. Removing the nuisance-score projection leaves $(P_\perp \bar{d})^\top \bar{\varepsilon}$, whose variance is (31). Expanding the projector by an SVD gives (32). $\square$

If instead $\nu \sim \mathcal{N}(m_\nu, \Gamma)$ with a proper prior, marginalization yields

$$r \mid a \sim \mathcal{N}(ad + Nm_\nu, \Sigma + N\Gamma N^\top), \quad (33)$$

$$I_{a|\nu}^{\text{Bayes}} = d^\top (\Sigma + N\Gamma N^\top)^{-1} d. \quad (34)$$

Woodbury's identity shows that (34) tends to the no-nuisance information as $\Gamma \rightarrow \mathbf{0}$ and to (31) as a full-rank nuisance

prior becomes diffuse. Thus zero deterministic efficient information is not a universal impossibility statement: a restrictive proper prior can still supply separation.

Corollary 1 (Pre-event information cannot reduce local event information): If pre-event data add nuisance precision $\Lambda_{\text{pre}} \succeq \mathbf{0}$, then

$$I_{a|\nu, \text{pre}} = I_{aa} - I_{a\nu}(I_{\nu\nu} + \Lambda_{\text{pre}})^{-1}I_{\nu a} \quad (35)$$

is nondecreasing in Λ_{pre} .

b: Justification.

If $\Lambda_2 \succeq \Lambda_1 \succeq \mathbf{0}$, matrix inversion reverses the order. Premultiplication and postmultiplication by the cross-information blocks preserve the positive-semidefinite order, and subtraction reverses it once more. $\square$

For a known source, known positive sign, and fixed covariance, the standardized matched statistic

$$T = \frac{d^\top \Sigma^{-1} r}{\sqrt{d^\top \Sigma^{-1} d}} \quad (36)$$

is standard normal under H_0 and has mean $a\sqrt{I}$ under amplitude a . A level- α one-sided test has power $1 - \Phi(z_{1-\alpha} - a\sqrt{I})$. Hence the amplitude required for power $1 - \beta$ is

$$a_{\alpha, \beta} = \frac{z_{1-\alpha} + z_{1-\beta}}{\sqrt{I}}. \quad (37)$$

This is a conditional known-source benchmark, not the threshold of the 137-support search. Its empirical use in the paper is limited to the exploratory source-wise detectability association.

E. PAIRWISE SOURCE GEOMETRY AND ONSET INFORMATION

Let two first-order source signatures be d_g and d_j under fixed precision $G \succeq \mathbf{0}$. Allow the competing source to refit its amplitude. The local profiled distance satisfies

$$b_*(a) = a \frac{d_j^\top G d_g}{d_j^\top G d_j} + O(a^2), \quad (38)$$

$$\Delta_{g \rightarrow j}^2(a) = a^2 J_{g \rightarrow j} + O(a^3), \quad (39)$$

$$J_{g \rightarrow j} = d_g^\top G d_g - \frac{(d_g^\top G d_j)^2}{d_j^\top G d_j} \geq 0. \quad (40)$$

Equation (40) is a pairwise local separation quantity, not the Fisher information for amplitudes conditional on the true support. For a multi-source support S , the latter is

$$I_S(a) = J_S(a)^\top G J_S(a), \quad J_S = \frac{\partial \mu_S}{\partial a_S}. \quad (41)$$

Positive definiteness of I_S permits local amplitude estimation conditional on S ; it does not exclude another support with the same mean.

Unknown onset has a separate degeneracy. For $\mu(a, \tau) = ad(\tau)$ and fixed covariance,

$$I(a, \tau) = \begin{bmatrix} d^\top \Sigma^{-1} d & ad^\top \Sigma^{-1} \dot{d} \\ a \dot{d}^\top \Sigma^{-1} d & a^2 \dot{d}^\top \Sigma^{-1} \dot{d} \end{bmatrix}. \quad (42)$$

Profiling amplitude by a Schur complement gives

$$I_{\tau|a} = a^2 \left[\dot{d}^\top \Sigma^{-1} \dot{d} - \frac{(d^\top \Sigma^{-1} \dot{d})^2}{d^\top \Sigma^{-1} d} \right]. \quad (43)$$

At $a = 0$, onset is unidentifiable. If $\dot{d}$ is proportional to d , delay and gain are locally confounded even for nonzero amplitude. The evaluated bank avoids this unresolved problem by fixing onset.

F. CANDIDATE INDISTINGUISHABILITY AND SPARSE INJECTIVITY

For candidate h and fixed integrity support S , define the affine mean set

$$\mathcal{M}_h(S) = s_{h,H} + \text{range}[\mathcal{O}_{h,H}, \mathcal{E}_{S,H}]. \quad (44)$$

The profiled margin in the main paper is half the squared distance between two such sets under R_H^{-1} .

Theorem 6 (Local candidate indistinguishability): If $D_{ij,S}(H) = 0$, the candidate manifolds intersect. At any intersection, the two candidates induce the same Gaussian law; no decision rule can distinguish them uniformly over the declared nuisance set.

a: Proof.

Zero distance between closed affine sets means that there exist nuisance vectors for which their means coincide. The covariance is common by the stated contract, so the two conditional probability measures are identical. For any decision region, the probabilities of choosing i and j under this common law sum to one; at least one conditional error is therefore at least one half. $\square$

For two fixed simple candidates with squared Mahalanobis distance $\Delta^2 = 2D_{ij,S}$ and equal priors, projection onto their mean difference reduces the optimal test to one dimension, giving

$$P_e^* = \Phi(-\Delta/2) = \Phi(-\sqrt{D_{ij,S}/2}). \quad (45)$$

This formula does not apply directly to amplitude-marginalized support mixtures.

For the local persistent-event model $Y_T = \mathcal{H}_T a + \varepsilon_T$, define the physically admissible difference set

$$\mathcal{D}_K = \{a - b : a, b \text{ admissible and } K\text{-sparse}\}. \quad (46)$$

Theorem 7 (Finite-horizon sparse event uniqueness): Every admissible K -sparse event is uniquely determined in the noise-free local model if and only if

$$\ker \mathcal{H}_T \cap \mathcal{D}_K = \{\mathbf{0}\}. \quad (47)$$

For a signed neighborhood in each support subspace, $\mathcal{D}_K = \mathcal{A}_{2K}$ locally, recovering the condition in the main paper.

b: Proof.

If two admissible events produce the same output, their nonzero difference belongs to both sets in (47). Conversely, any nonzero vector in the intersection is, by definition of $\mathcal{D}_K$, the difference of two admissible K -sparse events that produce the same output. $\square$

Unknown initial-state nuisance can be removed locally by whitening and projection:

$$\tilde{\mathcal{H}}_T = (I - P_{\text{range}(W_T \mathcal{O}_T)}) W_T \mathcal{H}_T, \quad W_T^T W_T = R_T^{-1}. \quad (48)$$

Theorem 7 then applies with $\tilde{\mathcal{H}}_T$. Define the restricted margin

$$\gamma_{2K}(T) = \min_{1 \leq |U| \leq 2K} \sigma_{\min}(\tilde{\mathcal{H}}_{T,U}). \quad (49)$$

If two K -sparse explanations both fit data within whitened residual ϵ , then

$$\|a - b\|_2 \leq \frac{2\epsilon}{\gamma_{2K}(T)}. \quad (50)$$

Indeed, the triangle inequality bounds $\|\tilde{\mathcal{H}}_T(a - b)\|_2$ by 2ϵ , while the definition of γ_{2K} gives the reverse lower bound. For the evaluated $K \leq 2$ bank, the relevant union size is four.

G. INFORMATION CANNOT DECREASE WHEN VALID MEASUREMENTS ARE ADDED

Partition the difference between two candidate means into old and added measurement blocks, $\delta = \text{col}(\delta_1, \delta_2)$, with joint covariance $\Sigma \succ 0$. Block Gaussian elimination gives

$$\delta^T \Sigma^{-1} \delta = \delta_1^T \Sigma_{11}^{-1} \delta_1 + \tilde{\delta}_2^T \Sigma_{2|1}^{-1} \tilde{\delta}_2, \quad (51)$$

$$\tilde{\delta}_2 = \delta_2 - \Sigma_{21} \Sigma_{11}^{-1} \delta_1, \quad (52)$$

$$\Sigma_{2|1} = \Sigma_{22} - \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}. \quad (53)$$

The conditional covariance is positive definite, so the second term is nonnegative. The inequality holds for each common nuisance vector; minimizing over the same nuisance class preserves it. Hence adding a valid PMU or later frame cannot reduce ideal profiled information. A concrete estimator may still degrade if its model, covariance, or optimization changes.

If whitened model errors under two supports satisfy $\|We_S\| \leq \epsilon_S$ and $\|We_R\| \leq \epsilon_R$, the reverse triangle inequality gives the robust separation bound

$$d_{\text{true}} \geq [d_{\text{nom}} - \epsilon_S - \epsilon_R]_+, \quad \mathcal{J}_{\text{true}} \geq \frac{1}{2} [d_{\text{nom}} - \epsilon_S - \epsilon_R]_+^2. \quad (54)$$

This is the required bridge for future simulator-shift claims; it is not evaluated by the current nominal load bank.

H. TEMPORAL INFORMATION UNDER THE FROZEN AR(1) CONTRACT

For true support S , competing support R , true amplitude a , and competitor amplitude b , let $\delta_t(b) = \mu_{S,t}(a) - \mu_{R,t}(b)$. The frozen temporal covariance is

$$\Sigma_T = R_T(\rho) \otimes \Omega, \quad [R_T]_{ij} = \rho^{|i-j|}, \quad |\rho| < 1. \quad (55)$$

For fixed b , its squared Mahalanobis separation has the innovation form

$$q_T(b) = \|\delta_1(b)\|_{\Omega^{-1}}^2 + \frac{1}{1 - \rho^2} \sum_{t=2}^T \|\delta_t(b) - \rho \delta_{t-1}(b)\|_{\Omega^{-1}}^2. \quad (56)$$

Define the profiled simple-model separation

$$\mathcal{J}_{S \rightarrow R}(T) = \frac{1}{2} \inf_{b \in \mathcal{A}_R} q_T(b). \quad (57)$$

It equals the minimum KL divergence over fixed-amplitude simple Gaussian competitors. It is not the KL divergence or Bayes factor of the amplitude-mixture support model.

Theorem 8 (Horizon monotonicity and persistent rate): If the amplitude domain and nuisance dimension do not change with T and all covariances are prefixes of the same process, then $\mathcal{J}_{S \rightarrow R}(T + 1) \geq \mathcal{J}_{S \rightarrow R}(T)$. If, in addition, $\delta_t(b)$ converges uniformly on a compact competitor domain to $\delta_\infty(b)$, then

$$\lim_{T \rightarrow \infty} \frac{2\mathcal{J}_{S \rightarrow R}(T)}{T} = \frac{1 - \rho}{1 + \rho} \inf_{b \in \mathcal{A}_R} \|\delta_\infty(b)\|_{\Omega^{-1}}^2. \quad (58)$$

a: Proof.

Equation (56) gives $q_{T+1}(b) = q_T(b) + s_{T+1}(b)$ with $s_{T+1} \geq 0$ for every fixed b . Taking the infimum proves monotonicity. Under uniform convergence, $\delta_t - \rho \delta_{t-1}$ converges uniformly to $(1 - \rho)\delta_\infty$. Divide (56) by T ; the first-frame term vanishes and the Cesàro mean converges uniformly to $(1 - \rho)^2(1 - \rho^2)^{-1} \|\delta_\infty\|_{\Omega^{-1}}^2$. Compactness and uniform convergence permit interchange of infimum and limit, giving (58). $\square$

If the profiled equilibrium distance is positive, separation grows as $\Theta(T)$. If an admissible competitor matches equilibrium and its innovation differences are square summable, evaluating (57) at that competitor gives a finite upper bound; monotonicity then implies convergence to a finite ceiling. Exact trajectory overlap gives zero separation for every horizon.

For a true pair $\{i, j\}$, a competing pair $\{i, k\}$ can refit the shared source. With whitened equilibrium columns $\bar{h}_i, \bar{h}_j, \bar{h}_k$,

$$a_j^2 \|(I - P_{i,k})\bar{h}_j\|_2^2 \leq a_j^2 \|(I - P_i)\bar{h}_j\|_2^2. \quad (59)$$

The inequality follows from $\text{span}(\bar{h}_i) \subseteq \text{span}(\bar{h}_i, \bar{h}_k)$. It formalizes why identifying which weak second source occurred cannot be easier than rejecting the shared single-source explanation.

I. HIGHER-ORDER CONTACT AND LOCAL DETECTION SCALE

First-order Fisher information can vanish while nonlinear separation remains. In whitened coordinates, let the no-event nuisance manifold and event curve be

$$F_0(b) = Nb + \frac{1}{2} H_0[b, b] + O(\|b\|^3), \quad (60)$$

$$c(a) = ad + a^2 q + O(a^3). \quad (61)$$

Theorem 9 (Relative first- and second-order separation): Assume the nuisance manifold has constant rank and use a regular chart after removing redundant coordinates. Define the profiled first-order coefficient

$$\gamma_1^2 = \|P_\perp d\|_2^2, \quad P_\perp = I - (WJ_c)(WJ_c)^\dagger, \quad (62)$$

where J_c is the nuisance tangent before whitening and d is the already whitened event tangent in (61). If $\gamma_1 > 0$, the local squared distance to the nuisance manifold is

$$\Delta^2(a) = a^2 \gamma_1^2 + O(a^3). \quad (63)$$

If $\gamma_1 = 0$, so the whitened event tangent lies in the nuisance tangent space, define

$$\kappa = P_\perp \left(q - \frac{1}{2} H_0[v, v] \right). \quad (64)$$

Then

$$\Delta^2(a) = a^4 \|\kappa\|_2^2 + O(a^5). \quad (65)$$

a: Proof.

The local minimizer satisfies $DF_0(b)^\top [F_0(b) - c(a)] = \mathbf{0}$. At the origin, its derivative with respect to the regular chart coordinate is $N^\top N \succ \mathbf{0}$, so the implicit-function theorem gives a smooth minimizing branch. Write $b_*(a) = av + a^2 w + O(a^3)$. At first order, $N^\top (Nv - d) = \mathbf{0}$, yielding residual $aP_\perp d + O(a^2)$ and (63). If $d = Nv$, the first-order residual vanishes. At second order, minimization projects $q - \frac{1}{2} H_0[v, v]$ onto range N ; the remaining normal component is (64). Squaring the residual gives (65). The projector is unchanged by any full-rank reparameterization of the same nuisance tangent space, so γ_1 is a property of the local manifolds and covariance metric rather than of a particular nuisance coordinate basis. $\square$

Equation (63) supplies a direct falsification test: for independently generated nonlinear trajectories with decreasing $|a|$, the profiled squared distance divided by a^2 should converge to γ_1^2 . That convergence study has not been run for the present ANDES bank; the retrospective distance association in the article is not a substitute for it.

If n independent Gaussian replications have local distance $\Delta(a) = |a|^r \|k_r\| + O(|a|^{r+1})$, their KL divergence to the closest null is $n\Delta^2(a)/2$. Pinsker's inequality bounds the power of any level- α test by

$$\text{Power}_a \leq \alpha + \min\{1, \sqrt{n}\Delta(a)/2\}. \quad (66)$$

Thus $n|a_n|^{2r} \rightarrow 0$ prevents power from separating from size. The reference scales are $n^{-1/2}$ for first-order contact and $n^{-1/4}$ for second-order contact; singular-information asymptotics are established more broadly [16]. Correlated frames in one PMU window are not independent replications and must be handled through their joint covariance.

All audited load directions in the present bank have nonzero first-order separation. The second-order result is therefore a boundary theorem, not the empirical explanation for weak-pair failure.

J. THEORY-TO-EVIDENCE BOUNDARY

TABLE 3. Computational closure of the simulator event semantics.

Check	Quantity	Acceptance pattern	Status
Right-limit consistency	$\ g_+(x(\tau + \epsilon), z(\tau + \epsilon), a)\ $	Solver-scale residual after re-equilibration; callback residual may be $O(a)$	Open
Algebraic derivatives	Analytic z_1, z_2 versus central finite differences	Agreement under a decreasing symmetric amplitude grid	Open
Taylor order	Constraint and state-approximation errors	Successive slopes 1, 2, 3, or amplitude-halving ratios 2, 4, 8	Open
Event families	Load, generation, and continuous branch-admittance perturbations	Same declared semantics and tolerances for all three	Open
Dynamic first order	$B_{r,a}$ and the retained PMU response	$\Delta y/a$ converges over the declared early-time window	Open
Dynamic second order	Analytic second variation versus nonlinear trajectories	Discrepancy below a preregistered tolerance	Open; discrepancy unresolved

TABLE 4. Status of the mathematical results relative to the frozen evidence.

Result	What it establishes	Present evidence boundary
Consistency-jump theorem	Differential-state continuity, algebraic re-equilibration, and direct versus dynamic PMU signatures	Mathematical DAE contract only; callback/right-limit logging and derivative convergence remain open
Functional equivalence and BLUE	Exact target recovery and minimum covariance in the finite-horizon local linear model	Nonlinear TEST accuracy is empirical; the threshold-sensitive structural audit is not a certificate
Efficient Fisher projection and γ_1	First-order amplitude information and local separation after declared nuisance is removed	Source-wise detectability association is exploratory; $\Delta^2(a)/a^2 \rightarrow \gamma_1^2$ has not been tested
Sparse injectivity and γ_{2k}	Noise-free uniqueness and a local stability constant for a fixed dictionary	T120 diagnostics use the frozen load bank; no general descriptor left-invertibility claim
Temporal rate theorem	Persistent equilibrium separation gives linear information growth; transient-only separation can saturate	Current long-horizon study is diagnostic and lacks independent operating-point validation
Relative-curvature theorem	Zero first-order information need not imply nonlinear indistinguishability	Curvature-dominated weak-event behavior was not observed in the frozen atlas
Robust separation bound	Nominal geometry survives only when its margin exceeds model error	Requires a future simulator-shift or independent-data contract

APPENDIX E ESTIMATOR AND LIKELIHOOD DETAILS

This appendix distinguishes two algorithms. Sections E-A–E-C specify the state estimator and load-support posterior that produced the reported results. Section E-F gives the mathematical design of the common state–event–integrity estimator; that final integration has not yet been evaluated.

A. BAYESIAN RECURSION AND THE STATE HIERARCHY

For a Markov state model, Bayesian filtering alternates prediction and correction [17]. More broadly, this construction treats the PMU task as a Bayesian inverse problem in which physical candidate models and latent states are compared through predictive evidence [18]:

$$p(x_k | y_{0:k-1}) = \int p(x_k | x_{k-1})p(x_{k-1} | y_{0:k-1}) dx_{k-1}, \quad (67)$$

$$p(x_k | y_{0:k}) = \frac{p(y_k | x_k)p(x_k | y_{0:k-1})}{\int p(y_k | x_k)p(x_k | y_{0:k-1}) dx_k}. \quad (68)$$

The denominator in (68) is the predictive evidence. Omitting it is harmless for a state mean under one fixed model, but not when physical candidates are compared.

B0 returns the equilibrium. B1 conditions the equilibrium prior on one PMU frame:

$$K_{\text{LMMSE}} = P_0 C^\top (C P_0 C^\top + R)^{-1}, \quad (69)$$

$$\hat{x}_k^{\text{B1}} = K_{\text{LMMSE}}(y_k - y_0). \quad (70)$$

B2 retains state and covariance between frames. For the linear Gaussian model, substituting Gaussian densities into (67)–

(68) yields

$$m_k^- = A_d m_{k-1}^+, \quad P_k^- = A_d P_{k-1}^+ A_d^\top + Q, \quad (71)$$

$$S_k = C P_k^- C^\top + R, \quad K_k = P_k^- C^\top S_k^{-1}, \quad (72)$$

$$m_k^+ = m_k^- + K_k(y_k - C m_k^-), \quad P_k^+ = (I - K_k C) P_k^-. \quad (73)$$

Theorem 10 (Linear-Gaussian reduction): Equations (71)–(73) are the exact posterior mean and covariance generated by the Bayesian recursion for the stated linear Gaussian model.

a: Proof.

An affine transformation of the Gaussian posterior at $k - 1$, followed by addition of independent Gaussian process noise, gives (71). The pair (x_k, y_k) is jointly Gaussian with cross covariance $P_k^- C^\top$ and measurement covariance S_k . The standard conditional-Gaussian formula gives (72)–(73). Equivalently, multiplying the predictive density by the likelihood and completing the square gives posterior precision $(P_k^-)^{-1} + C^\top R^{-1} C$; Woodbury's identity produces the same covariance and mean. $\square$

The implementation may use the Joseph covariance form

$$P_k^+ = (I - K_k C) P_k^- (I - K_k C)^\top + K_k R K_k^\top \quad (74)$$

to preserve symmetry and positive semidefiniteness numerically.

B3 uses the Rauch–Tung–Striebel backward recursion [19]:

$$J_t = P_t^+ A_d^\top (P_{t+1}^-)^{-1}, \quad (75)$$

$$m_{t|k} = m_t^+ + J_t(m_{t+1|k} - m_{t+1}^-), \quad (76)$$

$$P_{t|k} = P_t^+ + J_t(P_{t+1|k} - P_{t+1}^-)J_t^\top. \quad (77)$$

It estimates a past state using later measurements. The negative B3 result therefore diagnoses model mismatch; it does not imply that correct future information generally lacks value.

The nominal uncertainty correction rescales the rectangular voltage covariance as

$$\Sigma_{i,k}^{U2} = T \Sigma_{i,k} T^T, \quad T = \text{diag}(\sqrt{\tau_{\mathcal{R}}}, \sqrt{\tau_{\mathcal{S}}}), \quad (78)$$

with temperatures fitted on independent calibration trajectories. The observed-only adequacy statistic is a five-frame trailing mean of

$$a_k = (\nu_k - \mu_\nu)^T \Sigma_\nu^{-1} (\nu_k - \mu_\nu). \quad (79)$$

Neither operation changes the physical state estimate.

B. CONSTRAINED PHYSICAL RECENTERING

The static recentering estimator treats active/reactive load changes and active generator changes as nuisance coordinates d . It solves

$$(\hat{V}, \hat{d}) = \arg \min_{V,d} \|e_y(V)\|_2^2 + \gamma^2 \|g_{AC}(V, d)\|_2^2 + \lambda q_d^{-2} \|d\|_2^2. \quad (80)$$

Development data fix $\gamma = 10^3$, $q_d = 1$, and $\lambda = 0.01$. Because the AC constraint appears through a finite penalty rather than a normalized stochastic density, the implemented objective is a regularized physical estimator, not a fully normalized posterior.

For an exactly constrained local MAP, let $r(\chi)$ be the stacked residual, $c(\chi) = \mathbf{0}$ the equality constraints, and Ω the residual precision. A Gauss–Newton step solves

$$\begin{bmatrix} H & G_c^T \\ G_c & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Delta \chi \\ \lambda \end{bmatrix} = \begin{bmatrix} -\nabla J \\ -c \end{bmatrix}, \quad H = J_r^T \Omega J_r + H_{\text{prior}}. \quad (81)$$

Theorem 11 (Feasible local covariance): If G_c has full row rank and Z spans $\ker G_c$, then $Z^T H Z \succ \mathbf{0}$ makes the KKT matrix nonsingular. The local constrained Gaussian covariance is

$$P_\chi = Z(Z^T H Z)^{-1} Z^T. \quad (82)$$

a: Proof.

Any vector in the KKT nullspace satisfies $G_c v = \mathbf{0}$, hence $v = Zq$. Premultiplying the first block equation by v^T gives $q^T Z^T H Z q = 0$, so $q = \mathbf{0}$ and $v = \mathbf{0}$. Full row rank then forces the multiplier to zero. Parameterizing feasible perturbations by $\delta \chi = Zq$ gives Gaussian precision $Z^T H Z$ in q ; its linear pushforward is (82). $\square$

This covariance is local and unimodal. The reported joint NEES-like failure shows that it is not calibrated for the full shifted-plant experiment.

C. EVALUATED PHYSICAL LOAD-SUPPORT MODEL

Let r be the centered T -frame PMU window. For support S with amplitudes a_S , the evaluated mean is

$$\mu_S(a_S) = \sum_{i \in S} (a_i D_i + a_i^2 Q_i) + \sum_{\substack{i,j \in S \\ i < j}} a_i a_j Q_{ij}. \quad (83)$$

The W2 likelihood is

$$q_S(a_S) = \|r - \mu_S(a_S)\|_{\Sigma_T^{-1}}^2, \quad (84)$$

$$p(r | S, a_S) = \frac{\exp[-q_S(a_S)/2]}{(2\pi)^{n_y T/2} |\Sigma_T|^{1/2}}, \quad (85)$$

$$\Sigma_T = R_T(\rho) \otimes \Omega. \quad (86)$$

The covariance is fitted only on normal calibration records and remains fixed across supports. The selected priors are

$$p(K = 0, 1, 2) = (0.20, 0.50, 0.30), \quad (87)$$

$$p(S | K) = \binom{16}{K}^{-1}, \quad (88)$$

$$a_i | i \in S \sim \mathcal{N}(0, \sigma_a^2), \quad \sigma_a = 0.05. \quad (89)$$

For reference, the linear single-source model has a closed form.

Proposition 1 (Conjugate amplitude posterior and Bayes factor): Let $r = ad + \epsilon$, $\epsilon \sim \mathcal{N}(\mathbf{0}, \Sigma)$, and $a \sim \mathcal{N}(0, \sigma_a^2)$. Define $I = d^T \Sigma^{-1} d$ and $s = d^T \Sigma^{-1} r$. Then

$$v = (I + \sigma_a^{-2})^{-1}, \quad m = vs, \quad (90)$$

$$a | r, S \sim \mathcal{N}(m, v), \quad (91)$$

$$\log \frac{p(r | S)}{p(r | H_0)} = \frac{1}{2} \frac{s^2}{I + \sigma_a^{-2}} - \frac{1}{2} \log(1 + \sigma_a^2 I). \quad (92)$$

a: Proof.

Terms in the joint log density that depend on a are $-\frac{1}{2}[(I + \sigma_a^{-2})a^2 - 2sa]$. Completing the square gives mean m and variance v . Integrating the resulting Gaussian contributes $\sqrt{v/\sigma_a^2} \exp(vs^2/2)$ relative to H_0 ; taking its logarithm gives (92). The logarithmic term is the amplitude-volume or Occam penalty. $\square$

The quadratic multi-source mean is nonconjugate. Its evidence is

$$Z_S = p(r | S) = \int p(r | S, a_S) \prod_{i \in S} p(a_i) da_S. \quad (93)$$

With Q -node Gauss–Hermite abscissas x_q and weights w_q , the Gaussian-prior transformation $a_i = \sqrt{2}\sigma_a x_{q_i}$ gives

$$Z_S \approx \pi^{-|S|/2} \sum_{q_1=1}^Q \cdots \sum_{q_{|S|=1}^Q \left(\prod_{j=1}^{|S|} w_{q_j} \right) p(r | S, \sqrt{2}\sigma_a x_q). \quad (94)$$

The implementation uses $Q = 31$ and evaluates (94) in the log domain using log-sum-exp. Independent adaptive Gauss–Kronrod integration checks representative two-dimensional cases. These checks support numerical integration under the frozen model; they do not validate the physical model itself.

D. POSTERIOR OUTPUTS AND THE CARDINALITY-LOCALIZATION DISTINCTION

The normalized support posterior is

$$p(S | r) = \frac{Z_S p(S)}{\sum_{R \in \mathcal{H}} Z_R p(R)}. \quad (95)$$

It induces cardinality, source-inclusion, and amplitude distributions:

$$p(K = k | r) = \sum_{|S|=k} p(S | r), \quad (96)$$

$$p(i \in S | r) = \sum_{S: i \in S} p(S | r), \quad (97)$$

$$p(a_i | r) = p(i \notin S | r) \delta_0 + \sum_{S: i \in S} p(S | r) p(a_i | S, r). \quad (98)$$

Equation (98) is a model-averaged spike-and-slab output: source absence contributes a point mass at zero.

The experimentally observed gap between cardinality and support has an exact algebraic decomposition. Let $N_k = \binom{16}{k}$, $\ell_S = \log Z_S$, and $\ell_k^* = \max_{|S|=k} \ell_S$. Define the effective multiplicity

$$m_{\text{eff},k} = \sum_{|S|=k} \exp(\ell_S - \ell_k^*). \quad (99)$$

Proposition 2 (Exact effective-multiplicity identity): Under a uniform prior over supports of cardinality k ,

$$\log p(r | K = k) = \ell_k^* - \log N_k + \log m_{\text{eff},k}, \quad (100)$$

$$\max_{|S|=k} p(S | K = k, r) = m_{\text{eff},k}^{-1}. \quad (101)$$

α: Proof.

Factor $\exp(\ell_k^*)$ from $N_k^{-1} \sum_{|S|=k} \exp(\ell_S)$ and take logarithms to obtain (100). Conditional normalization within cardinality k divides the largest evidence by the same sum, giving (101). $\square$

Many similar pair supports can therefore increase evidence for $K = 2$ while preventing any one pair from dominating. Entropy separates the same two questions exactly because K is a deterministic function of S :

$$H(S | r) = H(K | r) + \sum_k p(K = k | r) H(S | K = k, r). \quad (102)$$

The first term measures cardinality uncertainty; the second measures localization ambiguity within each cardinality.

For set-valued output, sort supports by posterior mass and define the smallest credible set

$$C_q(r) = \arg \min_C |C| \quad \text{subject to} \quad \sum_{S \in C} p(S | r) \geq q. \quad (103)$$

Candidate-set size and coverage are more informative than an arbitrary Top- m when the posterior is multimodal. The present paper has not yet calibrated an abstention threshold or credible-set coverage prospectively.

E. STEP-BY-STEP EVALUATED BAYESIAN ALGORITHM

Table 5 records the exact order of the load-support computation. Steps marked “evaluated” produced the 137-hypothesis results; optional outputs are mathematically defined but are not promoted to validated claims.

The direct quadrature cost per record is

$$1 + GQ + \binom{G}{2} Q^2 \quad (104)$$

likelihood-node evaluations for $K \leq 2$. With $G = 16$ and $Q = 31$, this is 115,817 nodes before vectorization or reuse. The count explains why the support bank, integration order, and horizon must be frozen together when runtime is compared.

F. CANDIDATE-CONDITIONED JOINT STATE-EVENT DESIGN

The common-contract extension would infer state trajectory X , algebraic variables Z , event support S , amplitudes a_S , operating nuisance η , and integrity corruption b . Its posterior factorization is

$$p(X, Z, \eta, b, S, a_S | Y) \propto p(S) p(a_S | S) p(\chi_0) \times \prod_{t=0}^{T-1} p(\chi_{t+1} | \chi_t, S, a_S) \prod_{t=0}^T p(y_t | \chi_t, b_t, S, a_S), \quad (105)$$

subject to the candidate DAE constraints. Here χ_t denotes a consistent state/nuisance chart. A constrained negative twice-log density, excluding $p(S)$, is

$$\mathcal{J}_S^\circ = \|\chi_0 - \bar{\chi}_0\|_{P_0^{-1}}^2 + \sum_{t=0}^T \|r_{S,t}^{\text{pmu}}\|_{R_t^{-1}}^2 + \sum_{t=0}^{T-1} \|r_{S,t}^{\text{dyn}}\|_{Q_t^{-1}}^2 + 2\lambda_b \sum_{t=0}^T \|b_t\|_1. \quad (106)$$

The Laplace evidence must be evaluated in a common feasible measure. If $\hat{\xi}_S$ is a mode in local free coordinates of dimension d_S and H_S is the corresponding positive-definite Hessian, then

$$\log Z_S^{\text{Lap}} \approx -\frac{1}{2} \mathcal{J}_S^\circ(\hat{\xi}_S) + \frac{d_S}{2} \log(2\pi) - \frac{1}{2} \log \det H_S. \quad (107)$$

Chart Jacobians, prior normalizers, and candidate-dependent covariance determinants belong in $\mathcal{J}_S^\circ$ or the measure term. Comparing only optimized residuals drops the volume penalty and does not produce a posterior.

For a sliding window, eliminating old Gaussian variables creates an arrival prior. Write its canonical log kernel as $-\frac{1}{2} z^\top \Lambda z + h^\top z + c$. If the information matrix and vector are partitioned into variables a to eliminate and retained variables b , exact integration yields

$$\Lambda_{\text{new}} = \Lambda_{bb} - \Lambda_{ba} \Lambda_{aa}^{-1} \Lambda_{ab}, \quad (108)$$

$$h_{\text{new}} = h_b - \Lambda_{ba} \Lambda_{aa}^{-1} h_a, \quad (109)$$

$$c_{\text{new}} = c + \frac{1}{2} h_a^\top \Lambda_{aa}^{-1} h_a + \frac{n_a}{2} \log(2\pi) - \frac{1}{2} \log \det \Lambda_{aa}. \quad (110)$$

TABLE 5. Evaluated 137-hypothesis Bayesian load-support algorithm.

Step	Operation	Computation and frozen choice	Status
0	Declare contract	Fix 16 candidate load buses, $K \leq 2$, known onset, 30-frame window, eight PMUs, nominal topology, and no corrupted channels	Evaluated
1	Validate plant	Check equilibrium, terminal-current orientation, sampling, nonlinear solve acceptance, and first-order response before building the bank	Evaluated
2	Build physical dictionary	Generate centered D_i , self-curvature Q_i , and cross-interaction Q_{ij} from simulator-consistent interventions	Evaluated
3	Fit noise on CAL	Estimate channel covariance and AR(1) dependence from normal calibration records; freeze W2 before event TEST	Evaluated
4	Freeze priors and integrator	Set $p(K)$, uniform support priors, Gaussian amplitude slab, GH31 nodes/weights, and independent GK checks	Evaluated
5	Form the residual	Align the known onset, subtract the frozen normal mean, retain the fixed PMU/channel order, and apply the declared temporal covariance	Evaluated
6	Enumerate supports	Construct H_0 , 16 single-source supports, and 120 two-source supports; no learned source label prunes the bank	Evaluated
7	Integrate each support	Evaluate (93) with (94); retain quadrature-weighted amplitude moments	Evaluated
8	Normalize once	Add the support prior in log space, subtract the global log-sum-exp, and obtain (95)	Evaluated
9	Read decisions in order	Report event presence, $p(K r)$, conditional support ranking, source inclusion, and conditional amplitude intervals separately	Evaluated
10	Quantify ambiguity	Compute entropy, effective multiplicity, or C_q without relabeling them as validated abstention guarantees	Defined; partial evidence
11	Record provenance	Store the plant identifier, support, amplitude, noise seed, likelihood version, and frozen artifact hash for every independent record	Evaluated

The first two lines preserve the next MAP problem; the last line is also required to preserve evidence between candidates.

Table 6 gives the intended sequence. It is a specification for the next experiment, not a claim about the results reported in the main paper.

For a posterior state $\pi_t(S) = p(S | Y_{1:t})$, an idealized unit-cost support decision with per-frame delay cost c satisfies the Bellman equation

$$V(\pi) = \min \left\{ 1 - \max_S \pi(S), c + \mathbb{E}[V(\pi_{t+1}) | \pi_t = \pi] \right\}. \quad (111)$$

The one-step expected information gain is

$$I(S; y_{t+1} | Y_{1:t}) = H(S | Y_{1:t}) - \mathbb{E}[H(S | Y_{1:t+1})]. \quad (112)$$

Equations (111)–(112) motivate adaptive horizon selection. They are not an executed stopping rule for the current fixed-onset, fixed-window bank.

Finally, the hidden-voltage posterior must average across support uncertainty rather than report only the winning support. With $w_S = p(S | Y)$, conditional target mean m_S , and covariance P_S ,

$$\bar{m} = \sum_S w_S m_S, \quad (113)$$

$$\text{Cov}(\psi | Y) = \sum_S w_S P_S + \sum_S w_S (m_S - \bar{m})(m_S - \bar{m})^T. \quad (114)$$

The second term is between-support ambiguity. Dropping it would make the virtual-PMU field appear overconfident precisely when source hypotheses disagree.

G. ESTIMATOR CLAIM BOUNDARY

The executed state recursion is Bayesian under its local linear Gaussian model, and the load-source bank computes normalized Bayesian evidence under the frozen W2/L2 model. The physical recentering is a regularized constrained estimator with separately assessed uncertainty. The full candidate-conditioned state–event–integrity posterior, learned discrepancy, unknown-onset inference, changing-PMU operation,

and sequential stopping rule remain specified but unexecuted. No result in the main paper is attributed to those pending components.

APPENDIX F COMPLETE FROZEN EXPERIMENTAL CONTRACTS

The PowerDynamics validation ladder checks static translation, PMU terminal-current orientation, equilibrium residual, local modes, nonlinear acceptance, and first-order response before estimator scoring. All 980 main mismatch cases and 175 refinement cases pass the declared acceptance gates. The physical-recentering TEST includes 20 equilibria at each of three shift scales; a solve is accepted only when the optimizer succeeds, outputs are finite, and the maximum AC residual is at most 5×10^{-5} . Rejections remain in the denominator.

For GLOBAL-137, the physical dictionary, covariance, priors, onset, and quadrature are frozen before new trajectories are generated. Three disjoint noise realizations accompany each event path, and 100 additional H_0 records represent no event. The identifiers have zero overlap with 1,816 historical records. The T120 bank preserves exact prefixes at 30, 60, 90, and 120 frames; independent operating-point validation remains pending.

The ANDES ordinary split uses three replicas for training, one for development, and one for testing at each source configuration. The source-holdout audit removes the physical asset from every source-labeled operation and supplies the true event family at evaluation. These contracts answer interpolation and source generalization separately.

TABLE 6. Candidate-conditioned joint estimator design and current implementation boundary.

Step	Operation	Required calculation	Present status
1	Pre-event update	Use only pre-event measurements to update state and operating nuisance; retain cross covariance and normalized arrival evidence	State filter evaluated; joint nuisance pending
2	Integrity mask	Mark missing channels exactly and introduce sparse corruption variables for available but suspect channels	Ontology defined; common inference pending
3	Candidate generation	Enumerate physical family, support, onset, and severity domain without using source identity as a learned label	Load family evaluated; other families pending
4	Candidate prediction	Apply the candidate DAE/event map and the current PMU membership to generate a physical trajectory	Load bank evaluated
5	Candidate-conditioned solve	Minimize (106) on the feasible DAE manifold using the KKT system and continuation from the pre-event state	Not executed jointly
6	Learned discrepancy	If used, predict only candidate-residual discrepancy and covariance; do not replace the physical hypothesis or consume target source labels	Not implemented
7	Evidence normalization	Retain log determinants, prior densities, feasible-coordinate measures, and the Schur normalizer before comparing candidates	GH31 load evidence evaluated; joint Laplace evidence pending
8	Posterior outputs	Return state-functional mixture, cardinality, support, integrity state, severity, and candidate-set uncertainty	Separate state/load outputs only
9	Sequential decision	Continue observing when expected information gain justifies delay; use distinct thresholds for detection, cardinality, and exact source	Not evaluated
10	Reconfigure and verify	After accepting a physical change, rebuild the operating center and check subsequent innovations before declaring a persistent model update	Static recentering only

TABLE 7. Complete frozen-contract inventory. The main paper groups these contracts by research question.

Study	Frozen contract	Independent evaluation unit
Structural audit	Preregistered voltage/current target residual, horizons 0–180	31 hidden buses
Nominal reconstruction	20 DEV / 100 TEST; 91 frames at 30 Hz; B0–B3 frozen	Complete nonlinear trajectory
Nominal calibration	30 independent CAL trajectories; U1/U2 and discrepancy only	Complete nonlinear trajectory
Physical mismatch	Seven families and seven scales; B1/B2/U2 frozen	980 rebuilt plants
Breakpoint refinement	New seeds in transition bands; B2 frozen	175 rebuilt plants
Causal recentering	M1/M2/M6/M7; 160 DEV / 320 TEST; tune on DEV	Rebuilt plant trajectory
Regularized physical recentering	M6 PQ/PV/Slack basis; 30 DEV / 60 TEST; one snapshot	Rebuilt equilibrium
Load tangent audit	16 load buses; centered $\pm 0.5\%$ nonlinear simulations	Candidate-bus intervention
Bayesian load inference V1	Eight amplitudes; 2,560 event / 320 no-event TEST records	Source-amplitude-noise record
Bayesian load inference V2	1,000 normal CAL; 500 normal + 640 event DEV; weak and finite TEST blocks	Source-amplitude-noise record
Prospective 137-hypothesis inference	Fresh H_0 + 16 singles + 120 pairs; 3,488 paths; 10,564 decisions	Physical trajectory or complete record, as declared
Weak-regime mechanism audit	Frozen GH31 prefixes at 5–30 frames; 2,400 paths	Retrospective source-pair trajectory
Long-horizon diagnostic	Exact prefixes through 120 frames; 32 nominal pair paths; 137 hypotheses	Diagnostic trajectory, not population recovery
Known-source event diagnosis	Eight abnormal labels plus normal; 414 TRAIN / 138 DEV / 138 TEST	Complete ANDES trajectory
Source-transfer audit	Asset-disjoint folds; true family supplied	101 event parents, each evaluated across three seeds

APPENDIX G SECONDARY RESULTS

The main paper retains the figures needed for the scientific argument. This appendix records secondary diagnostics and exact values that would otherwise interrupt that argument. The two compact tables first preserve the exact mismatch and weak-signal values; the full-width panels then show the corresponding temporal, likelihood, and supervised-baseline diagnostics.

TABLE 8. Median B2 hidden-voltage HVRE at the largest mismatch scale, $m = 1.5$. Ratios use the 140-case main-grid nominal median of 0.014744%.

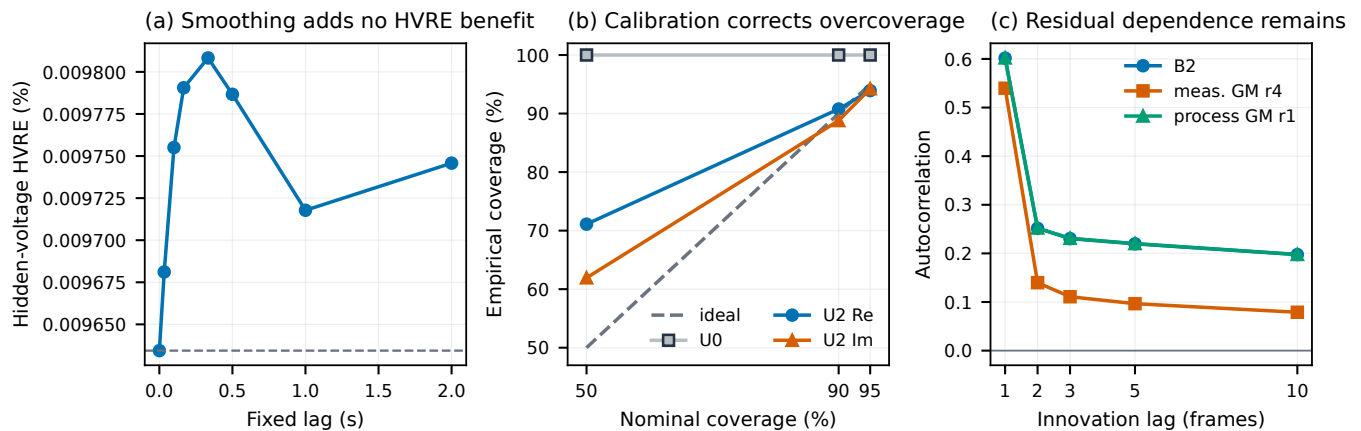
Mismatch family	HVRE (%)	Ratio
M1 network	0.250430	16.985×
M2 machine	0.014744	1.000×
M3 governor	0.014743	1.000×
M4 AVR	0.014744	1.000×
M5 load model	0.014761	1.001×
M6 operating point	0.122641	8.318×
M7 coupled	0.023615	1.602×

TABLE 9. Sign-pooled weak load-event TEST results for the frozen W2/L2 model. Each row contains 1,600 event cases (16 source buses, two signs, 50 noise replicas); the false-positive rate uses 800 no-event records.

$ a $ (%)	AUROC	FPR (%)	FNR (%)	Top-1 (%)	Top-3 (%)
0.010	0.733	0.25	83.1	13.0	37.1
0.020	0.906	0.25	45.0	31.8	69.0
0.050	0.995	0.25	4.5	73.4	95.9
0.100	1.000	0.25	0.2	93.2	99.4
0.200	1.000	0.25	0.0	98.9	100.0
0.400	1.000	0.25	0.0	100.0	100.0

TABLE 10. Frozen TEST uncertainty and colored-discrepancy ablations. U2 is fitted on a disjoint 30-trajectory calibration set; Gauss–Markov (GM) ranks are selected on that calibration set.

Variant	Hidden NLL	95% cov. Re (%)	95% cov. Im (%)	Innovation ACF(1)	HVRE (%)
U0 original	-13.477	100.0	100.0	0.602	0.009634
U2 Re/Im temperature	-16.184	93.9	94.2	0.602	0.009634
D2-M measurement GM r4	14.991	66.6	48.1	0.540	0.009788
D2-P process GM r1	-13.475	100.0	100.0	0.602	0.009638

**FIGURE 1.** Secondary temporal and uncertainty diagnostics for B2/B3 and calibration variants. These panels diagnose model adequacy; they do not upgrade nominal marginal coverage to a complete uncertainty claim.**TABLE 11.** Audited sequential event-diagnosis baseline on the known-source TEST split, mean over three fitted seeds. The independent split unit is the complete trajectory; timestamp-level scores retain the denominators of the frozen evaluator.

Configuration	Detection F1	Event macro-F1	Physical Top-1	Physical Top-3	Integrity Top-1	False alarms/min
136-feature hierarchy	0.965	0.791	0.496	0.646	0.870	4.01
448-feature hierarchy	0.975	0.815	0.590	0.717	0.865	1.45
448 features + availability	0.989	0.905	0.592	0.719	0.974	1.45

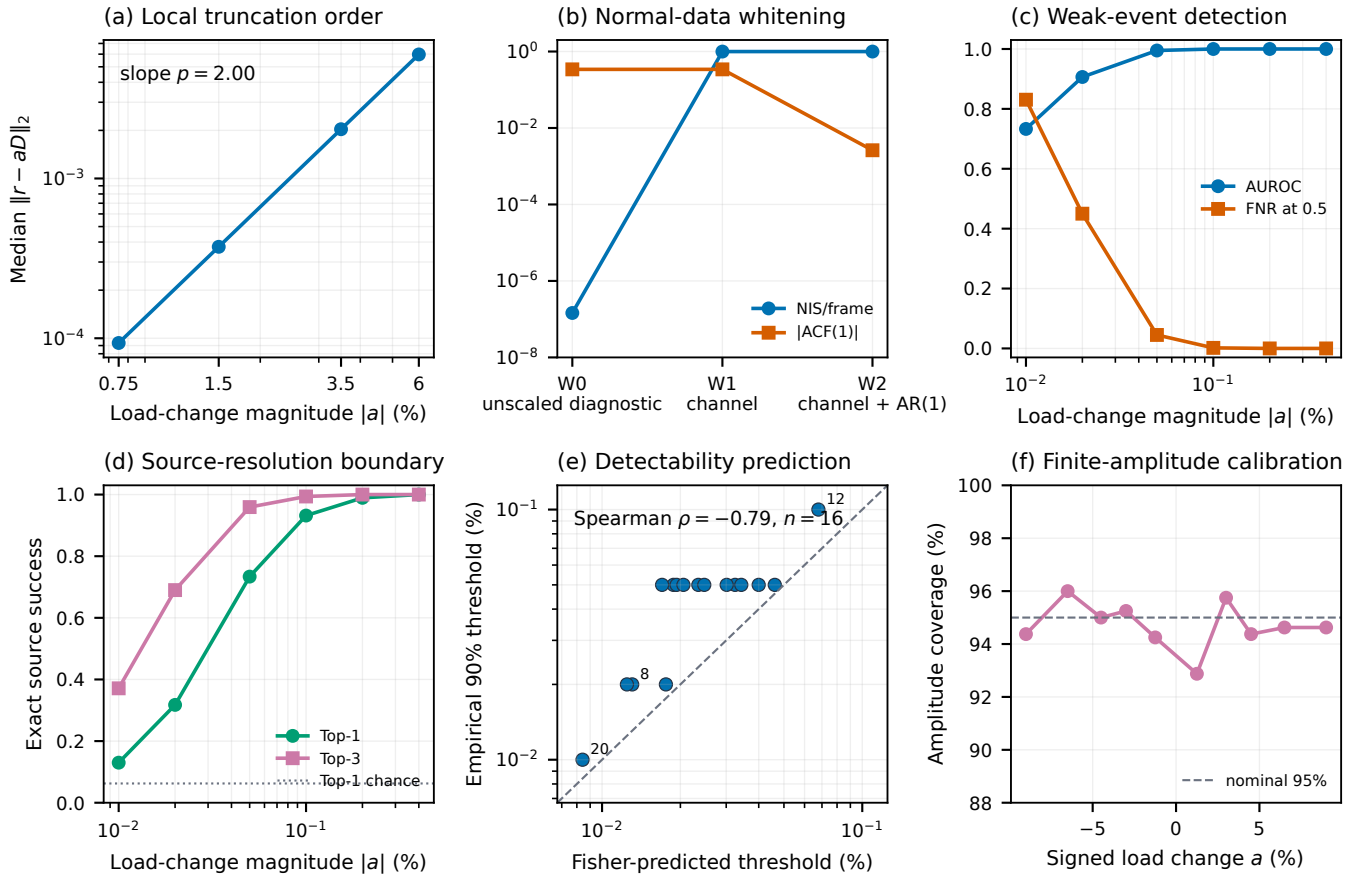


FIGURE 2. Single-source development evidence for the frozen load likelihood. The figure is retained as component validation; GLOBAL-137 in the main paper is the prospective support-recovery result.

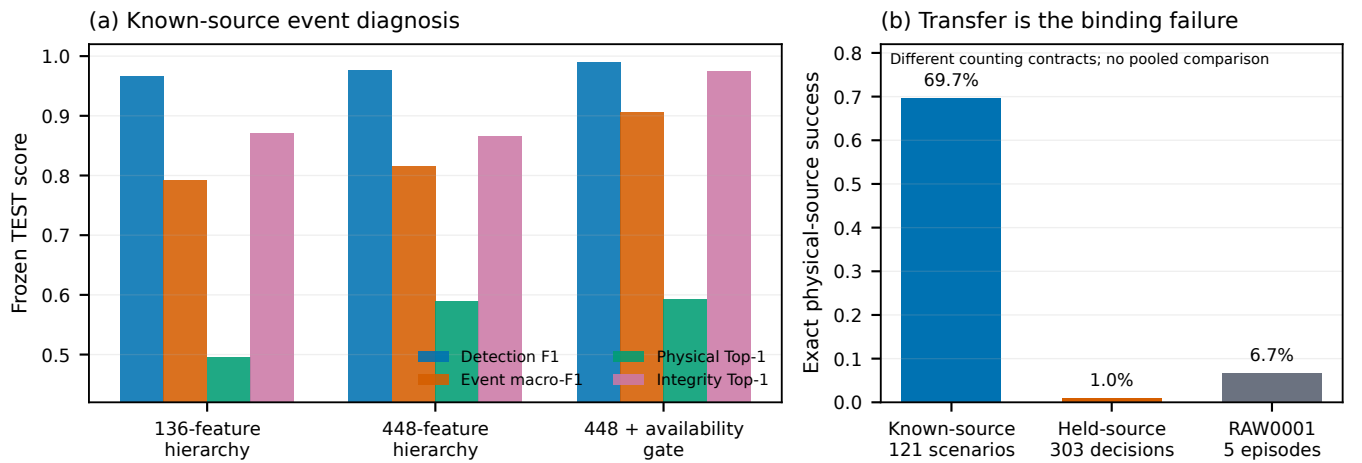


FIGURE 3. Secondary evidence for the supervised ANDES baseline, including known-source performance and the source-holdout failure.

REFERENCES

- [1] M. Katanic, J. Lygeros, and G. Hug, "Recursive dynamic state estimation for power systems with an incomplete nonlinear DAE model," *IET Generation, Transmission & Distribution*, vol. 18, no. 22, pp. 3657–3668, 2024.
- [2] S. A. Nugroho, A. F. Taha, N. Gatsis, and J. Zhao, "Observers for differential algebraic equation models of power networks: Jointly estimating dynamic and algebraic states," *IEEE Transactions on Control of Network Systems*, vol. 9, no. 3, pp. 1531–1543, 2022.
- [3] B. Qu, Z. Wang, B. Shen, H. Dong, and H. Liu, "Event-based joint state and unknown input estimation for energy networks: Handling multi-machine power grids," *IEEE Transactions on Network Science and Engineering*, vol. 10, no. 1, pp. 253–264, 2023.
- [4] Q.-H. Ngo, B. L. H. Nguyen, T. V. Vu, J. Zhang, and T. Ngo, "Physics-informed graphical neural network for power system state estimation," *Applied Energy*, vol. 358, p. 122602, 2024.
- [5] H. Yue, W. Zhang, U. C. Yilmaz, T. Yildiz, H. Huang, H. Liu, Y. Lin, and A. Abur, "Graph-learning-assisted state estimation using sparse heterogeneous measurements," *Electric Power Systems Research*, vol. 235, p. 110644, 2024.
- [6] S. Moshtaghi, B. Azimian, M. Golgol, and A. Pal, "Topology-aware graph neural network-based state estimation for PMU-unobservable power systems," *IEEE Transactions on Power Systems*, 2025.
- [7] S. Pandey, A. K. Srivastava, and B. G. Amidan, "A real time event detection, classification and localization using synchrophasor data," *IEEE Transactions on Power Systems*, vol. 35, no. 6, pp. 4421–4431, 2020.
- [8] T. Yildiz and A. Abur, "Convolutional neural network-assisted fault detection and location using few PMUs," *Electric Power Systems Research*, vol. 235, p. 110705, 2024.
- [9] Z. Li, J. Tian, and H. Liu, "A joint data and knowledge-driven method for power system disturbance localisation," *IET Generation, Transmission & Distribution*, vol. 18, no. 24, pp. 4078–4089, 2024.
- [10] A. Kummerow, M. Dirbas, C. Monsalve, and P. Bretschneider, "Siamese sigmoid networks for the open classification of grid disturbances in power transmission systems," *IET Smart Grid*, vol. 6, no. 2, pp. 136–146, 2023.
- [11] Y. Hu and Z. Cheng, "Generative AI for open-set event detection and classification in power systems using synchrophasor data," *IEEE Transactions on Industrial Informatics*, 2026, early access.
- [12] M. Rimkus, P. Kokoszka, D. Duan, X. Wang, and H. Wang, "Graph neural networks for the localization of faults in a partially observed regional transmission system," *Scandinavian Journal of Statistics*, vol. 52, no. 2, pp. 572–594, 2025.
- [13] Z.-M. Zhai, M. Moradi, and Y.-C. Lai, "Detecting attacks and estimating states of power grids from partial observations with machine learning," *PRX Energy*, vol. 4, no. 1, p. 013003, 2025.
- [14] S. M. Kay, *Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory*. Englewood Cliffs, NJ, USA: Prentice Hall, 1993.
- [15] A. N. Montanari, L. Freitas, D. Proverbio, and J. Gonçalves, "Functional observability and subspace reconstruction in nonlinear systems," *Physical Review Research*, vol. 4, no. 4, p. 043195, 2022.
- [16] A. Rotnitzky, D. R. Cox, M. Bottai, and J. M. Robins, "Likelihood-based inference with singular information matrix," *Bernoulli*, vol. 6, no. 2, pp. 243–284, 2000.
- [17] S. Särkkä, *Bayesian Filtering and Smoothing*. Cambridge, U.K.: Cambridge University Press, 2013.
- [18] A. M. Stuart, "Inverse problems: A bayesian perspective," *Acta Numerica*, vol. 19, pp. 451–559, 2010.
- [19] H. E. Rauch, F. Tung, and C. T. Striebel, "Maximum likelihood estimates of linear dynamic systems," *AIAA Journal*, vol. 3, no. 8, pp. 1445–1450, 1965.

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